MIT-WPU

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TECHNOLOGY, RESEARCH, SOCIAL INNOVATION & PARTNERSHIPS

School of Computer Science & Engineering

Department of Computer Science and Applications

2026–2027

SYNOPSIS

On

FORGE & FUEL MATRIX: Advanced Fitness, Workout Analytics & Intelligent Nutrition Tracking System

Mini Project

Course Code: XXXXXXXXX

FIRST YEAR MSC COMPUTER SCIENCE

Year: 2026–2027

Group Id: [GROUP ID]

Team Leader: [TEAM LEADER NAME]

S.No.	Team Members	PRN
1	[STUDENT NAME 1]	[PRN 1]
2	[STUDENT NAME 2]	[PRN 2]
3	[STUDENT NAME 3]	[PRN 3]
4	[STUDENT NAME 4]	[PRN 4]

Project Title: FORGE & FUEL MATRIX: Advanced Fitness, Workout Analytics & Intelligent Nutrition Tracking System

Name of the Mentor: [MENTOR NAME]

Declaration

This declaration is part of the Synopsis for project work entitled **"FORGE & FUEL MATRIX: Advanced Fitness, Workout Analytics & Intelligent Nutrition Tracking System"** submitted as part of academic requirement for Semester II of FIRST YEAR MSC COMPUTER SCIENCE to the Dr. Vishwanath Karad MIT World Peace University.

I, **[STUDENT NAME]** [**PRN**] solely declare that:

1. I will not use any unfair means to complete the project.
2. I will follow the discipline and the rules of the organization where I am doing the project.
3. I will not do or be part of any act which may impact the college reputation adversely.

The information I have given is true, complete and accurate. I understand that failure to give truthful, incomplete and inaccurate information may result in cancellation of my project work.

Date: [DD/MM/YYYY] **Name and Signature of Student**

Place: Pune

[STUDENT NAME]

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1. Brief Overview of the Project

FORGE & FUEL MATRIX is an all-in-one fitness and nutrition web application designed to help people manage their daily workouts, track their food intake, and learn proper exercise form in one simple, unified dashboard.

Purpose: Many beginners struggle because fitness tools are scattered across different apps (one for calorie counting, one for workouts, and videos for form). FORGE & FUEL solves this by combining scientific calorie calculations, animated exercise form guides, daily meal tracking, and motivational support into a single, clean web application.

Major Functions:

- **Automatic Calorie & Macro Calculator:** Calculates daily calorie needs (BMR & TDEE) and gives exact grams of Protein, Carbs, and Fats based on the user's goal (Fat Loss, Muscle Gain, or Maintenance).
- **Interactive Exercise Guide with 2D Visualizer:** Displays workouts by muscle group with an animated HTML5 Canvas visualizer showing the correct bar and movement path for lifts like Squats and Bench Press.
- **Daily Diet & Hydration Tracker:** Lets users search foods, log custom meals, track water intake, and view live progress bars toward their daily calorie limits.
- **Fitness Knowledge & Motivation:** Science-backed articles on nutrition and daily mindset quotes.
- **Theme Switcher:** 5 color themes (Cyber Neon, Sapphire Luxury, Crimson, Amber Gold, Arctic Light).

Target Users: College students, gym beginners, athletes, and personal trainers looking for a fast, free, and distraction-free fitness companion.

Expected Contribution: A lightweight, responsive web app that runs offline in any browser, requires no paid subscriptions, and keeps all personal fitness data safely on the user's device.

2. Background of the Project

Maintaining physical health is essential, yet most people find fitness apps frustrating. Commercial mobile apps frequently charge costly monthly subscriptions for basic features like macro counting, show excessive ads, and run slowly over poor gym internet connections. FORGE & FUEL was created to provide a modern, open, and fast web application that works completely in the browser without subscriptions or delays.

3. Domain / Technology Used

Project Domain: Web Application Development, Health Informatics & Sports Software.
Frontend Technologies:

- **HTML5:** Semantic structure for all pages and forms.
- **Vanilla CSS3:** Custom CSS properties (design tokens) for smooth layout, responsive mobile design, and 5 interactive color themes without heavy framework dependencies.
- **JavaScript (ES6 Modules):** Application routing, calorie calculation logic, and reactive UI updates.
- **HTML5 Canvas API:** Real-time 2D animated kinetic drawings demonstrating correct exercise form.
- **Vite 5:** Modern build tool for high-speed local development and optimized deployment.
- **HTML5 LocalStorage:** Client-side storage to save user profiles, logged meals, and workout sets permanently on the device without needing an external database server.

4. Problem Statement

Fitness beginners fail to reach their goals due to three common obstacles: (1) they do not know their daily caloric and protein targets, (2) they risk injuries because static workout images do not explain proper lifting form and movement cadence, and (3) commercial fitness apps are expensive and require continuous high-speed internet. **FORGE & FUEL MATRIX** directly solves this by providing a free, local-first web app that calculates exact dietary targets, visualizes exercise movement paths using animated 2D graphics, and logs daily meals and workouts with zero lag.

5. Objectives and Scope of the Project

5.1 Objectives

- **O1:** Implement the Mifflin-St Jeor scientific formula to accurately calculate Basal Metabolic Rate (BMR) and Total Daily Energy Expenditure (TDEE) based on user age, gender, height, weight, and activity level.
- **O2:** Build an automated macronutrient calculator that divides total calories into Protein, Carbs, and Fats according to the user's selected fitness goal (Cut, Bulk, or Recomposition).
- **O3:** Develop an animated HTML5 Canvas 2D engine that visually demonstrates correct exercise movement paths for key compound lifts.
- **O4:** Provide a real-time food and water tracking system where users can log meals, add custom items, and watch daily nutritional progress update instantly.
- **O5:** Create a responsive, 5-theme user interface using CSS design tokens that operates smoothly on both mobile phones and desktop computers.
- **O6:** Implement local-first data storage using browser LocalStorage so that all user profiles and logs are saved securely and work offline.

5.2 Scope

What the Project Covers: Complete biometric user profiling, automated calorie/macro calculations, interactive workout directory with muscle categorization, 2D animated form visualizer, meal and water logging, fitness knowledge articles, motivational quote bookmarks, and 5 color themes.

Exclusions (Future Enhancements): Bluetooth synchronization with wearable smartwatches (Apple Watch / Fitbit) and multi-device cloud synchronization will be incorporated in Phase 2.

6. Methodology / Implementation Plan

The project follows an **Iterative Agile Methodology**: (1) Formula research & mathematical engine, (2) UI layout & 5-theme token setup, (3) Exercise library with Canvas animations, (4) Diet tracker & LocalStorage integration, and (5) Testing and optimization.

6.1 System Design Diagrams

Figure 1: High-Level System Architecture

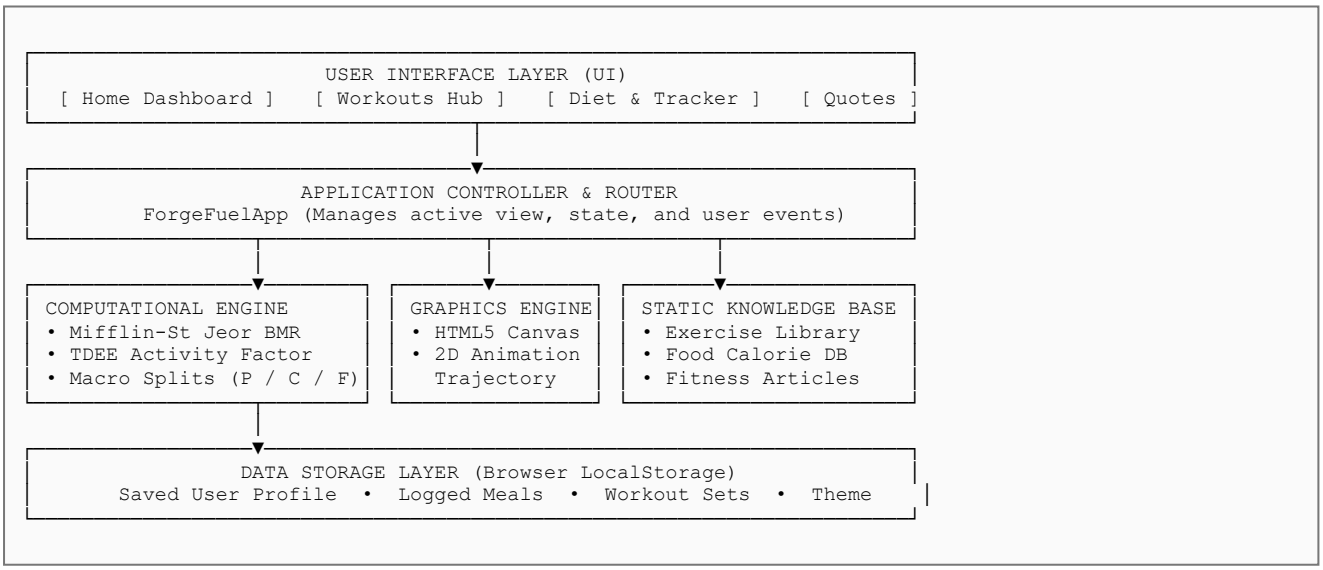
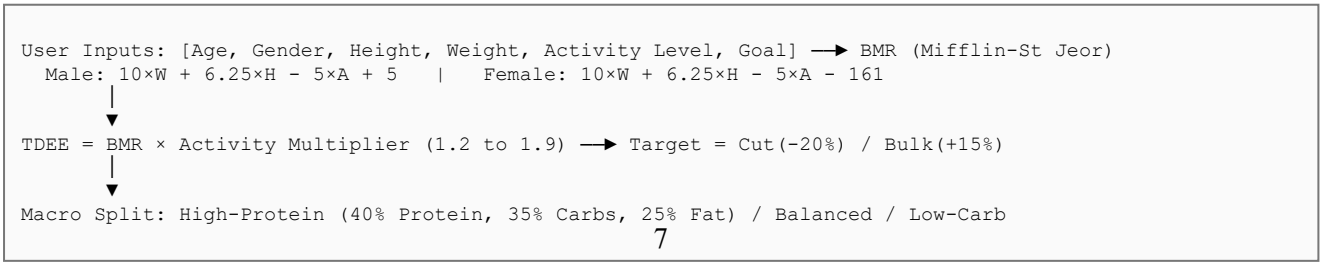
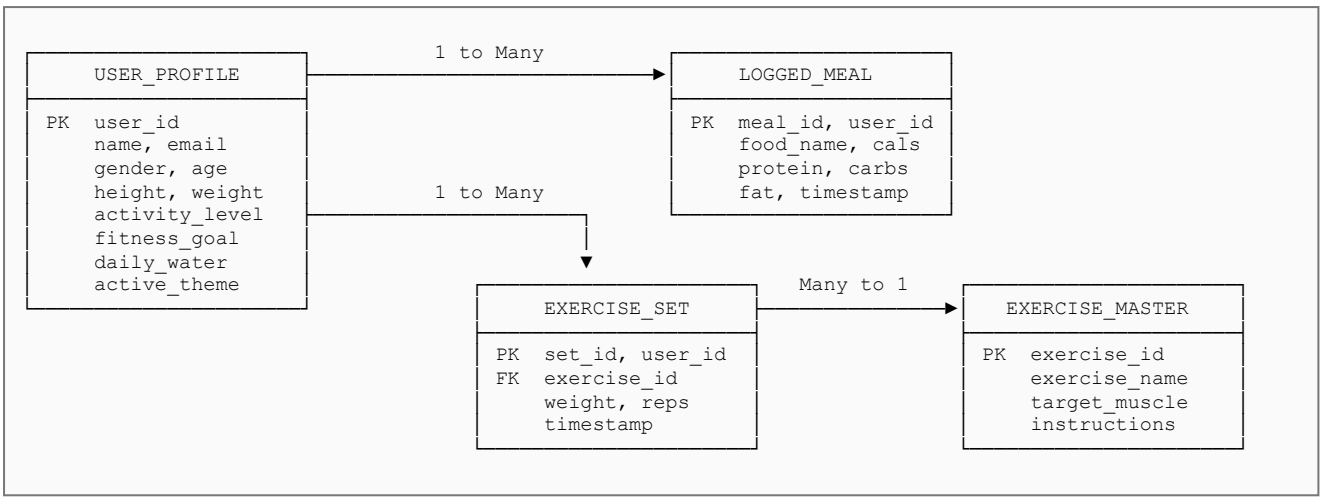


Figure 4: Calorie & Macro Calculation Flow



6.2 Entity-Relationship (ER) Diagram

Figure 5: Conceptual Entity-Relationship Data Schema



7. Expected Outcome

The project delivers a fully working, responsive web application that runs smoothly on any browser with the following outcomes:

- **Immediate Calorie & Macro Prescriptions:** Eliminates guesswork by providing exact daily caloric and macronutrient targets within seconds.
- **Injury Prevention via 2D Animations:** Visualizes movement paths directly in the browser so users can check proper exercise mechanics.
- **Effortless Daily Tracking:** Allows users to log food, water, and workout sets with instant visual progress bars.
- **Zero Cost & Complete Privacy:** Runs without paid subscriptions, keeping all user health data securely in the user's own browser storage.

8. Work Plan / Timeline Table

Sr. No.	Activity / Phase	Start Date	End Date	Status
1	Requirement Gathering & Formula Verification	01/08/2026	14/08/2026	Completed
2	Architecture Design & UI Theme Planning	15/08/2026	28/08/2026	Completed
3	Calorie & Macro Calculation Engine (BMR/TDEE)	29/08/2026	10/09/2026	Completed
4	Workout Hub & HTML5 Canvas 2D Visualizer	11/09/2026	22/09/2026	Completed
5	Diet Tracker, Food DB & LocalStorage Setup	23/09/2026	05/10/2026	Completed
6	Fitness Articles & Motivational Quotes Module	06/10/2026	18/10/2026	Completed
7	System Integration & Cross-Browser Testing	19/10/2026	31/10/2026	In Progress
8	Final Documentation & Project Synopsis Submission	01/11/2026	15/11/2026	In Progress